

BioNix_Week_7

Wednesday, 27 August 2025 14:14

- gc, garbage collection
- What is the Nix, the basic, and why we gonna use and learn in this internship
- Try to understand the Nix program whici written by previous intern
- Also could try to jump to BioNix

<https://github.com/PapenfussLab/bionix/tree/master>

<https://github.com/WEHI-ResearchComputing/BioNix-GettingStarted/wiki/5.-Virus-BreakEnd-Project:-Overview>

<https://github.com/PapenfussLab/gridss/blob/825bc8e1bb092e9bb2ccb0c1c51dc6e7e1a922dd/scripts/virusbreakend#L341>

Last login: Tue Sep 2 21:03:45 on ttys000

The default interactive shell is now zsh.

To update your account to use zsh, please run `chsh -s /bin/zsh`.

For more details, please visit <https://support.apple.com/kb/HT208050>.

MacBookPro:~ wesley\$ nix/

-bash: nix/: No such file or directory

MacBookPro:~ wesley\$ nix-shell

-bash: nix-shell: command not found

MacBookPro:~ wesley\$ # One-liner installer (multi-user "daemon" mode)

MacBookPro:~ wesley\$ curl -L <https://install.determinate.systems/nix> | sh -s -- install

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload	Upload	Total	Spent	Left
			Speed	Speed	Speed	Speed	Speed
0	0	0	0	0	--:--:--	--:--:--	0
100	18066	100	18066	0	0:00:01	0:00:01	29423

info: downloading installer https://install.determinate.systems/nix/tag/v3.9.1/nix-installer-x86_64-darwin

y

INFO nix-installer v3.9.1

`nix-installer` needs to run as `root`, attempting to escalate now via `sudo`...

Password:

Sorry, try again.

Password:

INFO nix-installer v3.9.1

INFO For a more robust Nix installation, use the Determinate package for macOS:

<https://dtr.mn/determinate-nix>

Install Determinate Nix?

Determinate Nix is tested and ready for macOS Tahoe when you are. It is built for people and teams, with dozens of user experience, performance, and reliability improvements. Selecting 'no' will install Nix from NixOS.

Proceed? ([Y]es/[n]o/[e]xplain): y

0: Planner error

1: Error executing action

2: Action `encrypt\_apfs\_volume` errored

3: The keychain lacks a password for the already existing "Nix Store" volume on disk `disk1`, consider removing the volume with `diskutil apfs deleteVolume "Nix Store"` (if you receive error -69888, you may need to run `sudo launchctl bootout system/org.nixos.darwin-store` and `sudo launchctl bootout system/org.nixos.nix-daemon` first)

Location:

/nix/var/nix/builds/nix-build-nix-installer-static-x86_64-apple-darwin-3.9.1.drv-1448-2621050073/nix-installer-source/src/cli/subcommand/install/mod.rs:180

Backtrace omitted. Run with RUST_BACKTRACE=1 environment variable to display it.

Run with RUST_BACKTRACE=full to include source snippets.

Consider reporting this error using this URL:

<https://github.com/DeterminateSystems/nix-installer/issues/new?title=%3Cautogenerated-issue%3E&body=%23%23+Error%0A%60%60%0AError%3A%0A+++0%3A+Planner+error%0A+++1%3A+Error+executing+action%0A+++2%3A+Action+%60encrypt+apfs+volume%60+errored%0A+++3%3A+The+keychain+lacks+a+password+for+the+already+existing+22Nix+Store%22+volume+on+disk+%60disk1%60%2C+consider+removing+the+volume+with+%60diskutil+apfs+deleteVolume+%22Nix+Store%22%60+%28if+you+receive+error+-69888%2C+you+may+need+to+run+%60sudo+launchctl+bootout+system%2Forg.nixos.darwin-store%60+and+%60sudo+launchctl+bootout+system%2Forg.nixos.nix-daemon%60+first%29%0A%60%60%60>

0A%0A%23%23+Metadata%0A%7Ckey%7Cvalue%7C%0A%7C--%7C--%7C%0A%7C**version**%7C3.9.1%7C%0A%7C**os**%7Cmacos%7C%0A%7C**arch**%7Cx86_64%7C%0A

WARN The Determinate Nix Installer failed.
Try our macOS-native package instead, which can handle almost anything:
<https://dtr.mn/determinate-nix>

```
MacBookPro:~ wesley$ . /nix/var/nix/profiles/default/etc/profile.d/nix-daemon.sh
-bash: /nix/var/nix/profiles/default/etc/profile.d/nix-daemon.sh: No such file or
directory
MacBookPro:~ wesley$ nix --version
-bash: nix: command not found
MacBookPro:~ wesley$ nix doctor
-bash: nix: command not found
MacBookPro:~ wesley$ # List APFS volumes and find the one named "Nix Store"
MacBookPro:~ wesley$ diskutil apfs list
APFS Container (1 found)
|
+-- Container disk1 FEC9F68A-5FF8-4C6F-8EE5-AFB30C17D8E3
=====
APFS Container Reference:      disk1
Size (Capacity Ceiling):      250685575168 B (250.7 GB)
Capacity In Use By Volumes:    215442087936 B (215.4 GB) (85.9% used)
Capacity Not Allocated:        35243487232 B (35.2 GB) (14.1% free)
|
+--< Physical Store disk0s2 64D25731-4C2A-4FB5-8D75-6C0FB288B175
-----
|
|   APFS Physical Store Disk:    disk0s2
|   Size:                        250685575168 B (250.7 GB)
|
+--> Volume disk1s1 EE9897DC-81D4-48A0-821C-F3301F24A78A
-----
|
|   APFS Volume Disk (Role):    disk1s1 (Data)
|   Name:                       Macintosh HD - Data (Case-insensitive)
|   Mount Point:                /System/Volumes/Data
|   Capacity Consumed:          191072038912 B (191.1 GB)
|   Sealed:                     No
|   FileVault:                 Yes (Unlocked)
|
+--> Volume disk1s2 BA19E2AB-8431-4CA9-82CE-9F9A176640F2
-----
|
|   APFS Volume Disk (Role):    disk1s2 (Preboot)
|   Name:                       Preboot (Case-insensitive)
|   Mount Point:                /System/Volumes/Preboot
|   Capacity Consumed:          2456018944 B (2.5 GB)
|   Sealed:                     No
|   FileVault:                 No
|
+--> Volume disk1s3 C48BD4A3-70F0-4F7F-9DD6-2FDF96DBBD5B
-----
|
|   APFS Volume Disk (Role):    disk1s3 (Recovery)
|   Name:                       Recovery (Case-insensitive)
|   Mount Point:                Not Mounted
|   Capacity Consumed:          1335275520 B (1.3 GB)
|   Sealed:                     No
|   FileVault:                 No
|
+--> Volume disk1s4 44C025B9-4024-49B4-B3C3-67235C41912E
-----
|
|   APFS Volume Disk (Role):    disk1s4 (VM)
|   Name:                       VM (Case-insensitive)
|   Mount Point:                /System/Volumes/VM
|   Capacity Consumed:          2147721216 B (2.1 GB)
|   Sealed:                     No
|   FileVault:                 No
|
+--> Volume disk1s5 35AB2D90-06A5-4994-B3E4-679AB7F2DEE9
-----
|
|   APFS Volume Disk (Role):    disk1s5 (System)
|   Name:                       Macintosh HD (Case-insensitive)
|   Mount Point:                Not Mounted
|   Capacity Consumed:          11262373888 B (11.3 GB)
|   Sealed:                     Yes
|   FileVault:                 Yes (Unlocked)
|   Encrypted:                 No
|
|   Snapshot:                   CB893573-A313-44ED-ACBC-C21B0CB1857C
|   Snapshot Disk:              disk1s5s1
|   Snapshot Mount Point:       /
|   Snapshot Sealed:            Yes
|
+--> Volume disk1s7 DAE99200-D657-435E-A697-EE98BEBA8CCC
-----
|
|   APFS Volume Disk (Role):    disk1s7 (No specific role)
```

```

Name: Nix Store (Case-insensitive)
Mount Point: /Volumes/Nix Store
Capacity Consumed: 7026372608 B (7.0 GB)
Sealed: No
FileVault: Yes (Unlocked)
MacBookPro:~ wesley$
MacBookPro:~ wesley$ # Delete it by name (simplest)
MacBookPro:~ wesley$ sudo diskutil apfs deleteVolume "Nix Store"
Started APFS operation
Deleting APFS Volume from its APFS Container
Unmounting disk1s7
Erasing any xART session referenced by DAE99200-D657-435E-A697-EE98BEBA8CCC
Deleting Volume
Removing any Preboot and Recovery Directories
Finished APFS operation
MacBookPro:~ wesley$ sudo diskutil apfs deleteVolume disk1s6
Started APFS operation
Deleting APFS Volume from its APFS Container
Unmounting disk1s6
The volume "Update" on disk1s6 couldn't be unmounted because it is in use by process
292 (com.apple.MobileSoftwareUpdate.CleanupPreparePathService)
Error: -69888: Couldn't unmount disk
MacBookPro:~ wesley$ sudo launchctl bootout system/org.nixos.darwin-store 2>/dev/null
|| true
MacBookPro:~ wesley$ sudo launchctl bootout system/org.nixos.nix-daemon 2>/dev/null ||
true
MacBookPro:~ wesley$ sudo diskutil apfs deleteVolume "Nix Store"
Could not find APFS Volume Nix Store
MacBookPro:~ wesley$ sudo rm -rf /nix 2>/dev/null || true
MacBookPro:~ wesley$ curl -L https://install.determinate.systems/nix | sh -s -- install
% Total % Received % Xferd Average Speed Time Time Time Current
0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0
100 18066 100 18066 0 0 17515 0 0:00:01 0:00:01 --:--:-- 58277
info: downloading installer https://install.determinate.systems/nix/tag/v3.9.1/nix-
installer-x86\_64-darwin
y
y
INFO nix-installer v3.9.1
`nix-installer` needs to run as `root`, attempting to escalate now via `sudo`...
INFO nix-installer v3.9.1
INFO For a more robust Nix installation, use the Determinate package for macOS:
https://dtr.mn/determinate-nix

```

Install Determinate Nix?

Determinate Nix is tested and ready for macOS Tahoe when you are. It is built for people and teams, with dozens of user experience, performance, and reliability improvements. Selecting 'no' will install Nix from NixOS.

Proceed? ([Y]es/[n]o/[e]xplain): Nix install plan (v3.9.1)
Planner: macOS

Configured settings:
* **determinate_nix:** true

Planned actions:

- * Install Determinate Nixd
- * Create an encrypted APFS volume `Nix Store` for Nix on `disk1` and add it to `/etc/fstab` mounting on `/nix`
- * Extract the bundled Nix (originally from /nix/store/0469inhbfqk1q8sd5ss6lj666gm3ksdf-nix-binary-tarball-3.9.1/nix-3.9.1-x86_64-darwin.tar.xz) to `/nix/temp-install-dir`
- * Create a directory tree in `/nix`
- * Synchronize /nix/var ownership
- * Move the downloaded Nix into `/nix`
- * Synchronize /nix/store ownership
- * Create build users (UID 351-382) and group (GID 350)
- * Configure Time Machine exclusions
- * Setup the default Nix profile
- * Place the Nix configuration in `/etc/nix/nix.conf`
- * Configure the shell profiles
- * Configuring zsh to support using Nix in non-interactive shells
- * Create a `launchctl` plist to put Nix into your PATH
- * Configure the Determinate Nix daemon
- * Remove directory `/nix/temp-install-dir`

Proceed? ([Y]es/[n]o/[e]xplain): **INFO** Step: Install Determinate Nixd
INFO Step: Create an encrypted APFS volume `Nix Store` for Nix on `disk1` and add it to `/etc/fstab` mounting on `/nix`
INFO Step: Provision Nix
INFO Step: Create build users (UID 351-382) and group (GID 350)
INFO Step: Configure Time Machine exclusions

```

INFO Step: Configure time machine exclusions
y INFO Step: Configure Nix
INFO Step: Configuring zsh to support using Nix in non-interactive shells
INFO Step: Create a `launchctl` plist to put Nix into your PATH
INFO Step: Configure the Determinate Nix daemon
INFO Step: Remove directory `/nix/temp-install-dir`
Nix was installed successfully!
To get started using Nix, open a new shell or run `
/nix/var/nix/profiles/default/etc/profile.d/nix-daemon.sh`

```

```

MacBookPro:~ wesley$ ./nix/var/nix/profiles/default/etc/profile.d/nix-daemon.sh
MacBookPro:~ wesley$ nix --version
nix (Determinate Nix 3.9.1) 2.30.2
MacBookPro:~ wesley$ nix doctor
warning: 'doctor' is a deprecated alias for 'config check'
[PASS] PATH contains only one nix version.
[PASS] All profiles are gcroots.
[PASS] Client protocol matches store protocol.
[INFO] You are not trusted by store uri: daemon
MacBookPro:~ wesley$ echo '. /nix/var/nix/profiles/default/etc/profile.d/nix-
daemon.sh' >> ~/.zshrc
MacBookPro:~ wesley$ git clone https://github.com/PapenfussLab/bionix.git
Cloning into 'bionix'...
remote: Enumerating objects: 2227, done.
remote: Counting objects: 100% (103/103), done.
remote: Compressing objects: 100% (78/78), done.
remote: Total 2227 (delta 49), reused 65 (delta 25), pack-reused 2124 (from 1)
Receiving objects: 100% (2227/2227), 3.05 MiB | 14.39 MiB/s, done.
Resolving deltas: 100% (1493/1493), done.
MacBookPro:~ wesley$ cd bionix
MacBookPro:bionix wesley$ ls -R examples || true
call.nix      ex-nextflow  ex-wdl       ref.fa       sample1-2.fq  sample2-2.fq
default.nix   ex-tnpair   README.md    sample1-1.fq  sample2-1.fq

examples/ex-nextflow:
nextflow-example1.nix

```

```

examples/ex-tnpair:
cluster.nix  default.nix  README.md    tnpair.nix

```

```

examples/ex-wdl:
wdl-scatter-gather.nix
MacBookPro:bionix wesley$ # If the repo exposes flake outputs:
MacBookPro:bionix wesley$ nix flake show || true
git+file:///Users/wesley/bionix?ref=refs/heads/master&rev=
34c73bcd175ad200cec79d7ca2e327c4ae691166
└─lib: unknown
MacBookPro:bionix wesley$ nix build .#examples || true
error: flake 'git+file:///Users/wesley/bionix' does not provide attribute 'packages.x86
_64-darwin.examples', 'legacyPackages.x86_64-darwin.examples' or 'examples'
MacBookPro:bionix wesley$
MacBookPro:bionix wesley$ # If there's no flake output for examples, use legacy nix:
MacBookPro:bionix wesley$ nix-build examples/default.nix
error:

```

```

... while calling the 'import' builtin
at /Users/wesley/bionix/examples/default.nix:3:12:
 2| # synthetic data in this directory.
 3| { bionix ? import <bionix> { } } :
  | ^
 4|

```

```

... while realising the context of a path

```

```

... while calling the 'findFile' builtin
at /Users/wesley/bionix/examples/default.nix:3:19:
 2| # synthetic data in this directory.
 3| { bionix ? import <bionix> { } } :
  | ^
 4|

```

```

error: file 'bionix' was not found in the Nix search path (add it using
$NIX_PATH or -I)
MacBookPro:bionix wesley$ cd ~/bionix
MacBookPro:bionix wesley$ export NIX_PATH="bionix=$PWD${NIX_PATH:+:$NIX_PATH}"
MacBookPro:bionix wesley$ nix-instantiate --eval -I bionix=$PWD --expr
'builtins.attrNames (import ./examples/default.nix {})'
error:

```

```

... while calling the 'attrNames' builtin
at <string>:1:1:
 1| builtins.attrNames (import ./examples/default.nix {})
  | ^

```

```

... while calling the 'import' builtin

```

```

... while calling the 'import' builtin
  at /Users/wesley/bionix/default.nix:1:13:
    1| { nixpkgs ? import <nixpkgs> { }, overlays ? [ ] }:
      | ^
    2|

```

(stack trace truncated; use '--show-trace' to show the full, detailed trace)

```

error: file 'nixpkgs' was not found in the Nix search path (add it using
$NIX_PATH or -I)
MacBookPro:bionix wesley$ nix-instantiate --eval \
> -I bionix=$PWD \
> -I nixpkgs=https://github.com/NixOS/nixpkgs/archive/nixos-24.05.tar.gz \
> --expr 'builtins.attrNames (import ./examples/default.nix {})'
unpacking 'https://github.com/NixOS/nixpkgs/archive/nixos-24.05.tar.gz' into the Git
cache...
error:

```

```

... while calling the 'attrNames' builtin
  at «string»:1:1:
    1| builtins.attrNames (import ./examples/default.nix {})
      | ^

```

```

... in the condition of the assert statement
  at /Users/wesley/bionix/tools/octopus-call.nix:38:5:
    37|     in
    38|     assert (length (unique refs) == 1);
      |     ^
    39|     stage {

```

(stack trace truncated; use '--show-trace' to show the full, detailed trace)

```

error: path '/nix/store/zr4n38cd3s5rnpb6ggh9r5yg1lhmal32-source.drv' is not
valid
MacBookPro:bionix wesley$ nix-instantiate --eval \
> -I bionix=$PWD \
> -I nixpkgs=https://github.com/NixOS/nixpkgs/archive/nixos-24.05.tar.gz \
> --expr 'builtins.attrNames (import ./examples/default.nix {})'
error:

```

```

... while calling the 'attrNames' builtin
  at «string»:1:1:
    1| builtins.attrNames (import ./examples/default.nix {})
      | ^

```

```

... in the condition of the assert statement
  at /Users/wesley/bionix/tools/octopus-call.nix:38:5:
    37|     in
    38|     assert (length (unique refs) == 1);
      |     ^
    39|     stage {

```

(stack trace truncated; use '--show-trace' to show the full, detailed trace)

```

error: path '/nix/store/zr4n38cd3s5rnpb6ggh9r5yg1lhmal32-source.drv' is not
valid
MacBookPro:bionix wesley$ # 1) go into the examples dir
MacBookPro:bionix wesley$ cd ~/bionix/examples
MacBookPro:examples wesley$
MacBookPro:examples wesley$ # 2) build the example (pin nixpkgs + point <bionix> to
parent)
MacBookPro:examples wesley$ nix build \
> -I bionix=./ \
> -I nixpkgs=https://github.com/NixOS/nixpkgs/archive/nixos-24.05.tar.gz \
> -f .
error: Cannot build '/nix/store/24bx209w0hmnxdjx7677nn4v199831b-bionix-bwa-mem.drv'.
Reason: builder failed with exit code 1.
Output paths:
/nix/store/dklwfd0nbgvbr3dzzb8ldfkkkg2fhsy23-bionix-bwa-mem
Last 25 log lines:
> -5 for split alignment, take the alignment with the smallest
query (not genomic) coordinate as primary
> -q don't modify mapQ of supplementary alignments
> -K INT process INT input bases in each batch regardless of
nThreads (for reproducibility) []
>
> -v INT verbosity level: 1=error, 2=warning, 3=message, 4
+=debugging [3]
> -T INT minimum score to output [30]
> -h INT[,INT] if there are <INT hits with score >80.00% of the max
score, output all in XA [5,200]
>
> -z FLOAT A second value may be given for alternate sequences.
The fraction of the max score to use with -h [0.800000].
specify the mean, standard deviation (10% of the mean if
absent). max

```

```

> -a          output all alignments for SE or unpaired PE
> -C          append FASTA/FASTQ comment to SAM output
> -V          output the reference FASTA header in the XR tag
> -Y          use soft clipping for supplementary alignments
> -M          mark shorter split hits as secondary
>
> -I FLOAT[,FLOAT[,INT[,INT]]]
>             specify the mean, standard deviation (10% of the mean if
absent), max
>             (4 sigma from the mean if absent) and min of the insert
size distribution.
>             FR orientation only. [inferred]
> -u          output XB instead of XA; XB is XA with the alignment
score and mapping quality added.
>
> Note: Please read the man page for detailed description of the command line
and options.
>
> [main_samview] fail to read the header from "-".
For full logs, run:
  nix log /nix/store/24bx209w0hmnxdjx76777nn4v199831b-bionix-bwa-mem.drv
error: Cannot build '/nix/store/8551l5zin8qpyjsf29gsj0bp0xllm39a-bionix-samtools-
sort.drv'.
Reason: 1 dependency failed.
Output paths:
  /nix/store/hw2nyp3byissdq99zyfcwiw3xhiafjm9-bionix-samtools-sort
error: Cannot build '/nix/store/q2ipvkp251xbv70mq5yixl8a48gdc9w-bionix-samtools-
fixmate.drv'.
Reason: 1 dependency failed.
Output paths:
  /nix/store/w65jkjppqcdr03dbjgs8iln07gysihnd-bionix-samtools-fixmate
error: Cannot build '/nix/store/5frq4vcvna0n536hwqkvq36wvd14rnlx-bionix-samtools-
sort.drv'.
Reason: 1 dependency failed.
Output paths:
  /nix/store/8rk157lz8cis652zbmdjdww0w1lc3hvn-bionix-samtools-sort
error: Cannot build '/nix/store/lzh062v69pszv9f84r7yf5nrac004gm0-bionix-samtools-
markdup.drv'.
Reason: 1 dependency failed.
Output paths:
  /nix/store/5d9zx0ndh98ppsmp9jaivfawzragf8kh-bionix-samtools-markdup
error: Cannot build '/nix/store/0fvzs4as2p5b08wmbihy9qfvaj7jj4y5-bionix-octopus-
call.drv'.
Reason: 1 dependency failed.
Output paths:
  /nix/store/hdazrfr9kjh09bg8wmkd2f08vc92m335-bionix-octopus-call
  /nix/store/mf7n97n9rahlvzwl508f0cc4mhp15zhn-bionix-octopus-call-evidence
MacBookPro:examples wesley$

```